

Axient: Debt-Free Finality for Leveraged Binary Event Markets

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Abstract

Leveraged event positions combine a loan that must be repaid with an outcome claim that may become non-tradable before oracle finality. The central design problem is therefore not terminal payoff accounting but pre-finality execution: a visible order book does not guarantee that quoted value will match, settle, and become available for debt repayment before the venue closes. This paper presents AXIENT, a physically backed margin layer for binary event markets that separates leverage maturity from claim maturity. At a hard-flat decision time the controller sells the smallest admissible quantity whose lower settled-proceeds envelope covers a policy-independent no-repayment upper bound on debt plus a registered buffer. We prove robust ex-ante debt clearing inside the operating set, pathwise debt extinction, maximal residual spot exposure, and invariance of lender credit-principal exposure to the final payout vector and dispute duration after debt is extinguished. We derive an exact book-dependent leverage envelope, path-consistent aggregate hard-flat conditions that evaluate proceeds and a policy-independent no-repayment debt-service bound on one common operating set, and scenario-conditional reserve bounds based on shared execution curves. An impossibility result shows that no backend-only mechanism with leverage above one can guarantee zero shortfall when closure, signer control, settlement, or executable liquidity may fail without independent collateral or an enforceable guarantee. A deterministic reference evaluation covers step books, partial fills, settlement delay, shared-book liquidation, zero liquidity, and payout vectors in $\{0, \frac{1}{2}, 1\}$. The results are conditional mechanism-design guarantees, not claims of calibrated venue performance or production safety.

Keywords: prediction markets; event contracts; leverage; margin lending; robust execution; conditional tokens; oracle disputes; liquidation.

JEL Classification: G13, G14, G18, G23.

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1 Introduction

Prediction markets translate beliefs about events into tradeable prices, but spot claims ordinarily contain no loan that must be repaid before resolution [Hanson, 2003, Wolfers and Zitzewitz, 2004, Manski, 2006]. Leverage adds a second clock. A trader may still hold an unresolved outcome claim while the lender already requires cash. This difference is especially consequential when matching and settlement are asynchronous, the actual close can precede the scheduled close, or the order book deteriorates under forced flow.

The paper studies a margin layer rather than a new prediction venue. A trader contributes collateral $C > 0$ and requests leverage $L \geq 1$. The acquisition budget and initial principal are

$$N = LC, \quad D_0 = (L - 1)C. \quad (1)$$

The combined budget purchases real collateralized outcome tokens. The financed leg is therefore a genuine loan, not a database-only notional. The mechanism must control the financed tokens, possess enforceable risk-reducing execution authority, and apply settled proceeds to debt before releasing residual value.

The key contribution is the separation of four values that are often conflated:

- (i) quoted book proceeds;
- (ii) matched proceeds;
- (iii) settled cash available to repay debt; and
- (iv) final redemption after the oracle payout vector is fixed.

Only the third object can extinguish the loan. The controller consequently makes its sale decision from a conservative lower envelope of *settled* proceeds, not from the displayed book or matched quantity.

The concise presentation retains the main model, theorem statements, proof arguments, and deterministic evidence. Extended derivations, registries, and implementation diagnostics remain available in the earlier arXiv version history.

Contributions.

- (C1) A timing and accounting model that distinguishes leverage maturity, oracle finality, and redemption.
- (C2) A robust auto-deleverage rule that maximizes residual funded exposure subject to debt clearance.
- (C3) Debt-free-finality and dispute-invariance theorems after confirmed repayment.
- (C4) An exact book-dependent leverage condition and a path-consistent aggregate shared-book certificate.
- (C5) A boundary theorem showing why backend instructions alone cannot create a universal zero-shortfall guarantee.
- (C6) A deterministic reference evaluation with explicit negative controls.

2 Relation to execution, robust optimization, and event-market design

The mechanism combines several literatures that normally study different balance-sheet objects. Prediction-market research studies information aggregation and price interpretation, but a quoted event probability is not itself a debt-repayment asset [Manski, 2006, Wolfers and Zitzewitz, 2004]. Conditional-token systems supply the physically backed payoff primitive, while optimistic or disputed oracle systems separate event occurrence from final payout [Gnosis, 2020, UMA Protocol, 2026].

Execution research treats a large order as a dynamic control problem rather than a mark-to-market calculation. Classical models balance expected cost against execution risk and market impact [Bertsimas and Lo, 1998, Almgren and Chriss, 2001]; limit-order-book models emphasize cancellations, discrete depth, and queue dynamics [Cont et al., 2010]. The Axient objective is narrower than optimal execution: it seeks the minimum admissible sale with a debt-clearing certificate, not the strategy that minimizes expected cost.

Robust optimization replaces a fully specified distribution with an explicit uncertainty set [Bertsimas and Sim, 2004]. That choice is deliberate. A Gaussian perturbation of cumulative depth can violate non-negativity, monotonicity, lot structure, and the discrete failures that dominate event venues. The operating set instead registers book transformations, fill fractions, fees, retry limits, settlement horizons, and control assumptions. A statistical or conformal lower bound may help propose and validate that set, but it is not silently substituted for the pathwise infimum or the policy-independent no-repayment debt bound in the theorem. Using a statistical bound directly would change the certificate into a probabilistic statement and would require the uncovered tail probability to enter admission and reserve policy.

The central distinction from a synthetic event perpetual is balance-sheet termination. A synthetic contract leaves a derivative obligation alive into the resolution zone. A physically backed margin layer can extinguish its loan before finality and leave only an ordinary outcome token. The two products can resemble one another before hard-flat, but they have different terminal liabilities.

3 Market, clocks, and balance sheet

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$. The filtration contains book observations, venue state, matches, settlement confirmations, balances, oracle messages, and chain events. No Gaussian, stationarity, or independence assumption is imposed on order-book evolution.

A binary event has complementary claims YES and NO and a final payout vector

$$\boldsymbol{\pi} = (\pi_{\text{YES}}, \pi_{\text{NO}}) \in [0, 1]^2, \quad \pi_{\text{YES}} + \pi_{\text{NO}} = 1. \quad (2)$$

Canonical outcomes are $(1, 0)$ and $(0, 1)$; voids may use another vector, including $(\frac{1}{2}, \frac{1}{2})$. Long-NO represents a short view on YES without unsecured negative inventory. Outcome-token collateral is economically distinct from the lender's loan receivable, as in collateralized conditional-token systems [Gnosis, 2020].

Let t_0 be position opening, u the hard-flat decision, ν the last relevant match, σ the first confirmed time debt is zero, T_c^{act} actual close, τ_f oracle finality, and τ_r redemption. The intended order is

$$t_0 < u \leq \nu \leq \sigma < T_c^{\text{act}} \leq \tau_f \leq \tau_r. \quad (3)$$

A scheduled close is only a planning input.

Definition 3.1 (Leverage and claim maturity). *Leverage maturity is σ , the first confirmed time at which settled proceeds have extinguished debt. Claim maturity is τ_f , and cash maturity is τ_r . In general $\sigma \leq \tau_f \leq \tau_r$.*

At time t , let Q_t be the held outcome-token quantity, K_t dedicated cash, and D_t debt. The financed balance sheet is identifiable:

$$\text{assets: } (Q_t, K_t), \quad \text{liability: } D_t. \quad (4)$$

A position is isolated by market and risk bucket; cross-margin is outside the base design.

Assumption 3.2 (Authority and lien). *While $D_t > 0$, an enforceable controller can submit and settle permitted risk-reducing orders, the borrower cannot remove or re-pledge the financed tokens or dedicated cash, and settled proceeds repay debt before residual value is released.*

Physical backing and non-custodial control are distinct. A controlled subaccount may be physically backed but custodial; a user-only wallet may be non-custodial yet unsuitable if the user can refuse a required liquidation signature.

4 Execution accounting

For an ordered bid ladder with prices p_j and sizes v_j , gross proceeds from selling x tokens are the step function

$$B_t^{\text{gross}}(x) = \sum_j p_j \min \left\{ v_j, \left[x - \sum_{k < j} v_k \right]^+ \right\}. \quad (5)$$

Fees, gas, partial fills, retries, and settlement failure act on this quantity. A match is not cash. For an execution policy Π and horizon Δ , let $B_{u,\Delta}^{\Pi}(x; \omega)$ be pathwise settled

proceeds. Debt service is bounded independently of the sale selected by the certificate. Let $r_t(\omega) \geq 0$ be the authorized debt rate and let $C_{u,t}^{\text{debt}}(\omega)$ be the non-decreasing authorized-charge process. The policy-independent no-repayment debt trajectory solves

$$\overline{D}_{u,u}^0(\omega) = D_u, \quad d\overline{D}_{u,t}^0(\omega) = r_t(\omega)\overline{D}_{u,t}^0(\omega)dt + dC_{u,t}^{\text{debt}}(\omega). \quad (6)$$

The corresponding policy-independent robust objects are

$$\underline{B}_{u,\Delta}^\Pi(x) = \inf_{\omega \in \mathcal{U}_u^{\text{op}}(\Delta)} B_{u,\Delta}^\Pi(x; \omega), \quad (7)$$

$$\overline{DS}_{u,\Delta}^0 = \sup_{\omega \in \mathcal{U}_u^{\text{op}}(\Delta)} \overline{D}_{u,u+\Delta}^0(\omega). \quad (8)$$

The superscript 0 denotes zero repayment inside the bound, not zero debt. Let $P_{u,t}^\Pi$ denote cumulative settlement-confirmed debt repayment from u through t under policy Π , and let $b_u^{\text{HF}} \geq 0$ be a versioned hard-flat buffer.

Definition 4.1 (Admissible debt-first policy). *On $[u, u + \Delta]$, an execution policy Π is admissible debt-first if: (i) only settlement-confirmed dedicated cash can reduce debt; (ii) every application is non-negative and capped at the debt outstanding immediately before application; (iii) cumulative repayment $P_{u,t}^\Pi$ is adapted, finite-variation, and non-decreasing; (iv) no new principal is originated after u , apart from the registered rate and authorized charges already represented in (6); (v) one settlement-evidence identity can be consumed at most once; and (vi) residual cash is released only after the debt-first application.*

Proposition 4.2 (No-repayment domination). *For every policy satisfying Definition 4.1, cumulative repayment through t plus remaining debt at t is no greater than $\overline{D}_{u,t}^0$ on the same path:*

$$P_{u,t}^\Pi + D_t^\Pi \leq \overline{D}_{u,t}^0.$$

Hence (8) is independent of the candidate sale x and upper-bounds total debt service for every admissible execution policy.

Proof. Set $H_{u,t}^\Pi = P_{u,t}^\Pi + D_t^\Pi$. A capped debt application transfers the same amount from D^Π to P^Π , so it does not increase H^Π . Between applications,

$$dH_{u,t}^\Pi = r_t D_t^\Pi dt + dC_{u,t}^{\text{debt}} \leq r_t H_{u,t}^\Pi dt + dC_{u,t}^{\text{debt}},$$

where $D_t^\Pi \leq H_{u,t}^\Pi$ and $r_t \geq 0$. Both processes start at D_u . Comparison with (6) gives $H_{u,t}^\Pi \leq \overline{D}_{u,t}^0$. The outstanding-debt cap in Definition 4.1 is load-bearing: without it, cumulative recorded repayment could exceed the no-repayment trajectory after debt reaches zero. $\square$

The operating set contains bounded adverse book transformations, partial fills, fees, and settlement delays that remain inside the registered horizon. A broader stress set

$\mathcal{U}^{\text{stress}}$ records named adverse scenarios but does not inherit the certificate. Closure before settlement, zero executable bids, signer loss, collateral escape, persistent settlement failure, or aggregate flow beyond shared-book capacity are failure paths.

Let $\mathcal{Q}_u(Q) = \{0 = y_0 < \dots < y_m \leq Q\}$ be the admissible lot grid. The robust feasible set and planned sale are

$$\hat{\mathcal{F}}_{u,\Delta}(Q) = \{x \in \mathcal{Q}_u(Q) : K_u + \underline{B}_{u,\Delta}^\Pi(x) \geq \overline{\text{DS}}_{u,\Delta}^0 + b_u^{\text{HF}}\}, \quad (9)$$

$$\hat{x}_u = \min \hat{\mathcal{F}}_{u,\Delta}(Q). \quad (10)$$

If the set is empty, the position is not certified for ordinary hard-flat.

After settlement, let $B_\sigma^{\text{audit}}(x)$ be cumulative confirmed proceeds and $D_{\sigma-}$ debt immediately before application. The realized audit minimum is

$$x_\sigma^* = \min\{x : K_u + B_\sigma^{\text{audit}}(x) \geq D_{\sigma-}\}. \quad (11)$$

It explains realized overshoot but is never used circularly to select the order.

Proposition 4.3 (Minimality and residual exposure). *If $\hat{x}_u$ exists, no smaller admissible quantity has the same robust certificate. If the realized audit minimum exists, it leaves the largest residual funded exposure $Q - x_\sigma^*$ among all realized debt-clearing sales.*

The incremental accounting for a settled fill $(\Delta x_j, \Delta b_j)$ is

$$p_j = \min\{D_j, K_j + \Delta b_j\}, \quad (12)$$

$$Q_{j+1} = Q_j - \Delta x_j, \quad D_{j+1} = D_j - p_j, \quad K_{j+1} = K_j + \Delta b_j - p_j. \quad (13)$$

This update instantiates Definition 4.1: debt never becomes negative, over-application is rejected by the cap, and already-applied cash is never counted twice.

5 Debt clearing and debt-free finality

Theorem 5.1 (Robust ex-ante debt clearing). *Fix u, Δ, Π , and $\mathcal{U}_u^{\text{op}}(\Delta)$. Suppose $\hat{x}_u$ exists, the realized path remains in that operating set, actual close occurs after $u + \Delta$, authority and lien remain enforceable, every settlement delay admitted by the operating set remains within Δ , and the controller follows an admissible debt-first policy in Definition 4.1. Then the controller can stop no later than the certified cap and reaches a finite $\sigma \leq u + \Delta$ with*

$$D_\sigma = 0, \quad x_\sigma^* \leq \hat{x}_u. \quad (14)$$

Proof sketch. At the planned cap, the lower settled-proceeds envelope plus dedicated cash covers the no-repayment debt-service envelope and buffer. On every path in the same operating set, realized settled proceeds are at least the lower envelope. Proposition 4.2 bounds cumulative repayment plus residual debt by the policy-independent envelope. Therefore confirmed cash clears debt no later than the cap. Minimality gives the inequality for the audit quantity. $\square$

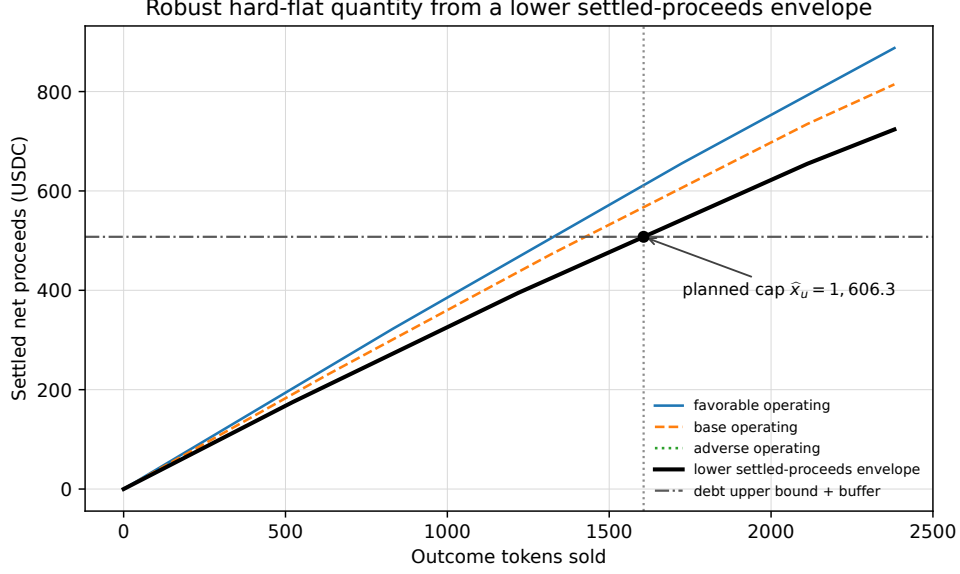


Figure 1: The ex-ante sale is selected from the lower settled-proceeds envelope. The realized audit minimum may be smaller on favorable paths.

After debt application, any remaining token quantity is fully funded spot exposure. Define lender credit-principal loss through the extinguished loan by

$$\mathcal{L}^{\text{loan}} = \left[D_{\tau_r^-} - K_{\tau_r^-}^{\text{loan}} - \text{loan recoveries} \right]^+. \quad (15)$$

Theorem 5.2 (Debt-free finality). *If debt has been extinguished at $\sigma < \infty$, no new borrowing occurs, residual tokens are not re-pledged, and later redemption is paid from outcome collateral separate from the loan receivable, then for every admissible payout vector and every dispute or redemption duration,*

$$\mathcal{L}^{\text{loan}} = 0. \quad (16)$$

The final payout changes the trader's residual claim value, not lender principal through the extinguished loan.

Proof sketch. Debt is zero after σ and cannot reappear without a new borrowing event. Residual outcome tokens are no longer pledged to the extinguished loan. Hence later oracle and redemption states do not create a new loan receivable or shortfall through that channel. $\square$

Corollary 5.3 (Dispute-duration invariance). *Conditional on confirmed debt extinction, extending the interval $\tau_f - \sigma$ does not change lender credit-principal exposure through the extinguished loan.*

The result is role-specific. Operators may remain exposed to custody, service-level, legal, reputational, oracle-governance, or redemption risks.

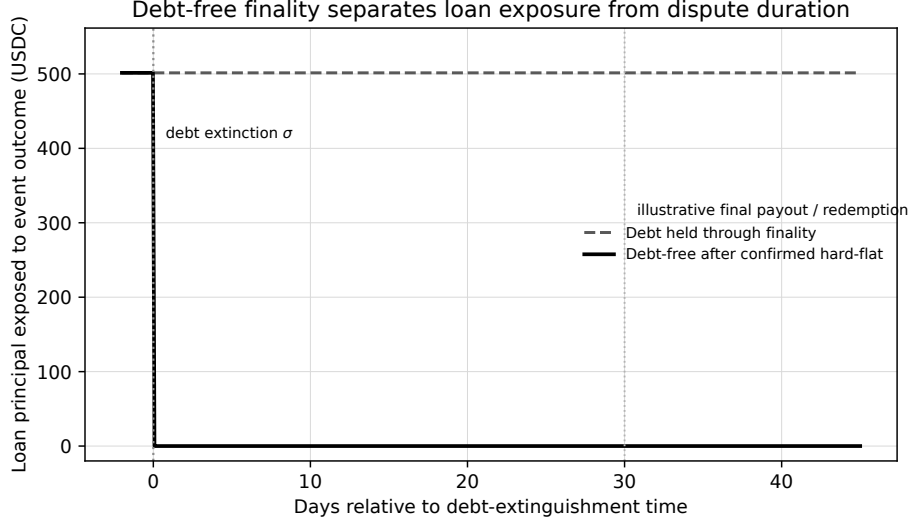


Figure 2: Once settled proceeds extinguish debt, the lender’s loan channel is invariant to the residual token payout and dispute duration.

Proposition 5.4 (Positive-debt impossibility). *Suppose $D_t > K_t$, the held token may pay zero, and there is no independent guarantee. Then some admissible payout vector yields positive lender shortfall if no debt-clearing sale occurs before finality.*

This proposition explains why leverage maturity must precede claim maturity unless additional collateral or guarantees cover the debt.

6 Leverage and aggregate liquidity

For leverage L , let the entry ask curve determine acquired quantity $Q_0(L)$ and residual cash $K_0(L)$. Let $\overline{DS}^0(L)$ be the no-repayment debt-service envelope and $b^{\text{HF}}(L)$ the hard-flat buffer. Define the exact robust surplus

$$\Psi(L) = K_0(L) + \underline{B}_{u,\Delta}^{\Pi}(Q_0(L)) - \overline{DS}^0(L) - b^{\text{HF}}(L). \quad (17)$$

Proposition 6.1 (Exact book-dependent leverage condition). *A leverage tier is fully-sale debt-clearing under the registered operating set if and only if $\Psi(L) \geq 0$. On a finite policy grid, the maximum exact tier is the largest admissible L satisfying this inequality.*

If net recovery is linear, gN , and fees are omitted, the benchmark condition becomes

$$gLC \geq (L - 1)C \iff g \geq 1 - \frac{1}{L}. \quad (18)$$

For $2\times$, $3\times$, and $5\times$, the corresponding recovery thresholds are 50%, 66.7%, and 80% before extra buffers. This scalar formula is intuition only; actual certification uses entry and exit ladders.

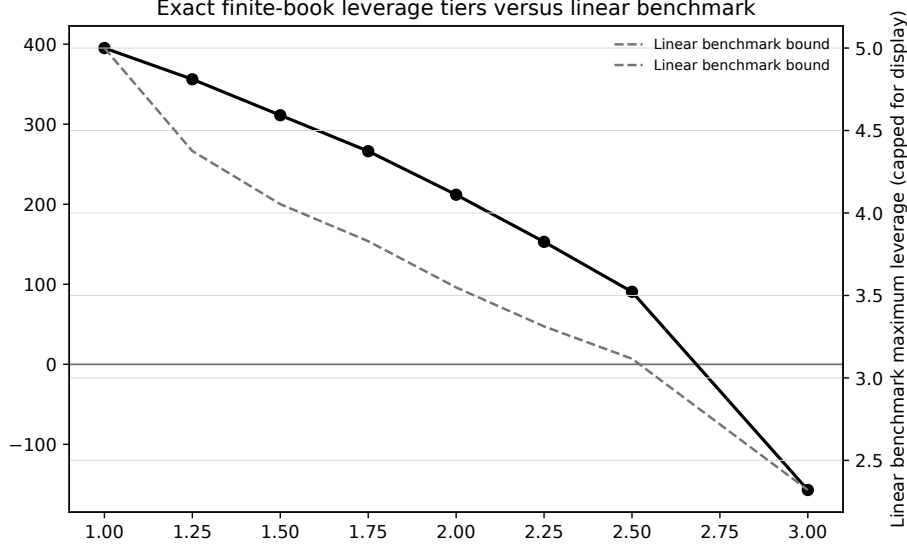


Figure 3: Illustrative exact leverage envelope from discrete entry asks and a conservative exit curve.

Multiple positions cannot each reuse the same top-of-book depth. For a cluster $\mathcal{C}$, let $\mathbf{x} = (x_i)_{i \in \mathcal{C}}$ be planned sales and let $\mathcal{U}_{\mathcal{C},u}^{\text{op}}(\Delta)$ be one aggregate operating set containing the shared-book, correlated settlement, and debt-service paths used by the pooled certificate. Define

$$\underline{B}_{u,\Delta}^{\Pi,\text{agg}}(\mathbf{x}) = \inf_{\omega \in \mathcal{U}_{\mathcal{C},u}^{\text{op}}(\Delta)} B_{u,\Delta}^{\Pi,\text{agg}}(\mathbf{x}; \omega), \quad (19)$$

$$\overline{\text{DS}}_{u,\Delta}^{0,\text{agg}} = \sup_{\omega \in \mathcal{U}_{\mathcal{C},u}^{\text{op}}(\Delta)} \sum_{i \in \mathcal{C}} \overline{D}_{i,u,u+\Delta}^0(\omega). \quad (20)$$

Proposition 6.2 (Top-of-book double counting). *Unless the book is perfectly linear over total size, the sum of proceeds obtained by re-anchoring the same book independently for every position weakly exceeds the proceeds from executing the combined quantity, and is strictly larger when the combined order reaches worse levels.*

Theorem 6.3 (Path-consistent pooled debt clearing). *Suppose the realized cluster path remains in $\mathcal{U}_{\mathcal{C},u}^{\text{op}}(\Delta)$ and the authority, non-escape, settlement-horizon, and no-new-debt conditions of the single-position certificate hold for every position. If*

$$\sum_{i \in \mathcal{C}} K_i + \underline{B}_{u,\Delta}^{\Pi,\text{agg}}(\mathbf{x}) \geq \overline{\text{DS}}_{u,\Delta}^{0,\text{agg}} + b_{\mathcal{C},u}^{\text{agg}}, \quad (21)$$

then pooled settled cash clears aggregate recognized debt service with buffer $b_{\mathcal{C},u}^{\text{agg}}$ on every path in that operating set.

Proof. On the realized aggregate path, shared-book settled proceeds are bounded below by (19), while Proposition 4.2 and (20) bound aggregate debt service above by the sum of the no-repayment trajectories on that same path. Substitution into (21) gives

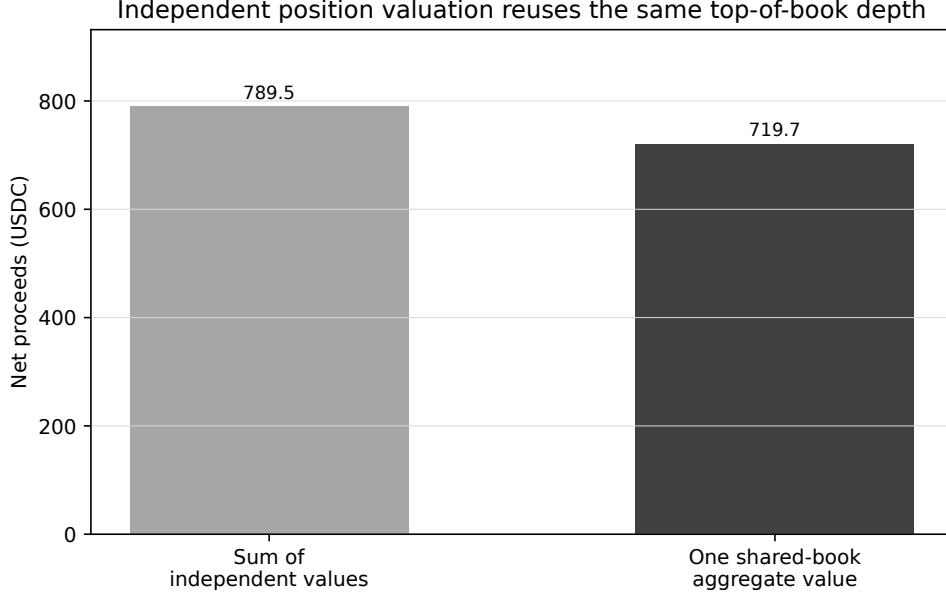


Figure 4: Independent position estimates reuse the same favorable depth; aggregate execution consumes the shared book once.

non-negative pooled surplus after debt service and buffer. Both bounds refer to the same path domain, so no independently selected worst paths are combined. $\square$

A separable sufficient bound is available only with an explicit domain condition. If individual debt-service envelopes are defined on $\mathcal{U}_{i,u}^{\text{op}}(\Delta)$ and

$$\mathcal{U}_{\mathcal{C},u}^{\text{op}}(\Delta) \subseteq \bigcap_{i \in \mathcal{C}} \mathcal{U}_{i,u}^{\text{op}}(\Delta), \quad (22)$$

then

$$\overline{\text{DS}}_{u,\Delta}^{0,\text{agg}} \leq \sum_{i \in \mathcal{C}} \overline{\text{DS}}_{i,u,\Delta}^0. \quad (23)$$

Without (22), summing positionwise envelopes does not certify an aggregate path. Priority, if needed, is frozen before execution rather than chosen after observing losses.

7 Failure boundary and reserves

Theorem 7.1 (No universal backend-only zero-shortfall guarantee). *Consider leverage $L > 1$ financed only by user collateral and outcome tokens, with no independent collateral, enforceable liquidity put, or third-party guarantee. If admissible paths include market closure before settlement, zero settled proceeds, loss of execution authority, or collateral escape while uncovered debt is positive, then zero lender shortfall cannot be guaranteed over all payout vectors and paths.*

The theorem does not prohibit operation. It requires the protocol to distinguish a certified operating region from externally capitalized tail states.

Table 1: Illustrative robust sale results.

Path	Audit minimum	Planned cap	Result
Favorable operating	1,311.2140	1,606.2949	debt cleared
Base operating	1,409.2884	1,606.2949	debt cleared
Adverse operating	1,589.1399	1,606.2949	debt cleared
Thin depth	–	–	shortfall 299.2882
Zero liquidity	–	–	shortfall 502.6907

For a risk cluster k and scenario s , define aggregate shortfall

$$\ell_k(s) = \left[\sum_{i \in \mathcal{C}_k} D_i(s) - \sum_{i \in \mathcal{C}_k} K_i(s) - \mathcal{B}_k^{(s)}(\mathbf{Q}_k) - G_k(s) \right]^+. \quad (24)$$

A segregated reserve $\mathcal{R}$ covers every scenario in a finite registered set $\mathcal{S}$ if

$$\mathcal{R} \geq \sup_{s \in \mathcal{S}} \sum_k \ell_k(s). \quad (25)$$

This is scenario-conditional, not universal. Reserve use must be reported separately from ordinary hard-flat success.

If available reserve R is smaller than attributed residual shortfall ℓ , a transparent allocation is

$$r_i = \begin{cases} \ell_i, & \sum_j \ell_j \leq R, \\ R\ell_i / \sum_j \ell_j, & \sum_j \ell_j > R. \end{cases} \quad (26)$$

It preserves non-negativity, caps each payment by the corresponding loss, and allocates exactly $\min\{R, \sum_i \ell_i\}$.

8 Deterministic evaluation

The reference evaluation uses finite step books, discrete lots, three operating paths, named stress paths, and settlement horizons of 5, 30, and 120 seconds. Parameters are illustrative and author-specified; they are not measurements from a venue.

The planned cap clears all operating paths while preserving more residual exposure on favorable paths. Negative controls remain negative: thin depth, zero liquidity, premature close, signer loss, collateral escape, and persistent settlement failure are not relabelled as robust success. Aggregate fixtures show that independently re-anchored liquidation values exceed one shared-book execution value. Finality fixtures evaluate $\pi_h \in \{0, \frac{1}{2}, 1\}$ and dispute delays from zero to 365 days; loan-channel loss remains zero after debt extinction.

Deterministic verification establishes code/formula consistency on registered fixtures. It does not estimate the probability of the operating set, market-maker response, settlement latency, signer enforceability, or commercial liquidity.

Table 2: Core lifecycle states and admissible transitions.

State	Permitted actions	Exit condition
OPEN	Risk-increasing and reducing orders within limits	Risk trigger, user close, or reduce-only threshold
REDUCE_ONLY	Cancel/replace, sell, repay, reconcile	Debt cleared or hard-flat deadline reached
HARD_FLAT	Registered risk-reducing execution only	Settled proceeds clear debt or failure path is declared
PARTIALLY SETTLED	Retry settlement; reduce debt by the confirmed amount	Full debt clearance, timeout, or conflict
DEBT_FREE SPOT	Hold or redeem the residual funded claim	Final payout and redemption
SHORTFALL	Reserve/backstop and incident handling	Loss allocated and position reconciled
DISPUTED / DELAYED	No new loan exposure; monitor payout evidence	Final vector or permanent failure classification

9 Position state machine and monitoring

A robust certificate is useful only if the runtime state machine preserves its assumptions. The reference lifecycle separates economic and operational states rather than compressing everything into ‘open’ and ‘closed’.

Monitoring must distinguish risk signals from accounting evidence. A stale or gapped book blocks new leverage but does not itself alter debt. A match updates pending execution but not cash. A settlement confirmation updates cash and debt. An oracle proposal updates expected claim value but does not finalize redemption. This separation prevents a single optimistic status flag from controlling multiple balance-sheet transitions.

The controller tracks at least the following quantities at market, account, and aggregate levels:

- age, sequence, and hash of the book used for the certificate;
- planned sale, matched quantity, settled quantity, and remaining certified capacity;
- debt principal, accrued interest, admitted fees, and dedicated cash;
- actual-close indicators and time remaining under the registered horizon;
- aggregate quantity already reserved against the shared execution curve;
- signer, lien, venue, RPC, and settlement capabilities;
- pending evidence identifiers and replay/conflict status; and
- reserve and backstop availability, reported separately from ordinary feasibility.

A risk trigger can move the position to reduce-only before hard-flat. Examples include declining robust surplus $\Psi(L)$, concentration growth, settlement backlog, capability expiry, or approach to the close buffer. The policy should prefer early risk reduction over waiting for a formal liquidation threshold because event-market liquidity can disappear discontinuously.

When ordinary execution fails, the system enters a named failure state rather than continuing to display the robust certificate. This is a substantive safety property: the protocol must expose that it has left the modeled region, preserve the evidence needed for reserve allocation, and prevent risk-increasing actions while recovery is attempted.

10 Implementation boundary

A reference architecture contains:

- (1) a market and venue-capability registry;
- (2) isolated collateral and position accounting;
- (3) a debt ledger with accrued interest and settlement-confirmed repayment;
- (4) a risk engine that evaluates entry and aggregate exit capacity;
- (5) a hard-flat executor with delegated authority;
- (6) a settlement-evidence registry with replay and conflict controls;
- (7) a fee and residual router; and
- (8) a segregated reserve and incident state machine.

The financial source of truth should be on-chain or otherwise independently auditable; market data, quote construction, routing, and keeper processes may remain replaceable off-chain agents. Venue capabilities are heterogeneous. A venue that exposes public books but no enforceable lien or delegated liquidation right cannot support the same guarantee as a contract-controlled vault.

The base mechanism is fail closed: no risk-increasing transition after reduce-only, no transition into finality states while debt remains positive, no debt reduction from a match, and no release of financed collateral before repayment. Security review must cover access control, evidence replay, integer rounding, fee ordering, early close, chain reorganization, signer compromise, and aggregate depth reuse.

11 Limitations and conclusion

The most important limitation is empirical. The operating set, lower settled-proceeds envelope, and stress transformations are author-specified. A robust theorem can be correct yet economically unhelpful if the set is narrow or selected after outcomes. Production use requires prospective calibration, a disjoint evaluation window, and public reporting of every breach. Statistical or conformal coverage can validate a proposed

operating policy, but it does not upgrade a probabilistic bound into the deterministic infimum used by the certificate.

The mechanism does not solve venue suspension, oracle manipulation, informed trading, legal characterization, custody risk, or market-maker strategic response. It does not prove optimal execution or trader welfare. A reserve is only as credible as the scenario catalogue, shared-book model, asset segregation, and legal availability. The result is also conditional on enforceable control over the financed position.

Within those boundaries, debt-free finality provides a sharp design principle: a leveraged event position should cease to be a loan before it becomes an unresolved finality claim. Robust settled-proceeds planning determines whether that transition is feasible; aggregate execution prevents depth reuse; and confirmed repayment converts the remaining claim into ordinary fully funded spot exposure. The final oracle outcome can then alter the trader’s payoff without reopening lender principal through the extinguished loan.

12 Controller certificate and audit checklist

The implementation-facing decision can be summarized by the following sequence.

- Step 1:** Freeze the position quantity, debt, dedicated cash, venue capability state, lot grid, operating set, settlement horizon, and versioned buffer.
- Step 2:** Construct the lower settled-proceeds envelope over the operating paths and the debt-service upper envelope on the same operating paths.
- Step 3:** Find the smallest admissible quantity satisfying (10). If none exists, reject ordinary hard-flat and enter a named shortfall or backstop state.
- Step 4:** Submit only risk-reducing orders, preserving correlation identifiers and distinguishing submitted, matched, settled, and failed amounts.
- Step 5:** After each settlement confirmation, apply (13); stop new submissions when debt is zero, and classify later settlement as explicit overshoot.
- Step 6:** Reconstruct the realized audit minimum (11), residual token quantity, aggregate depth consumption, and final accounting state.

A valid certificate records more than a number. It includes the input snapshot hash, operating-set version, execution-policy version, proceeds, debt-service, and buffer bounds, lot grid, planned quantity, actual-close status, signer/lien capability, settlement evidence identifiers, and aggregate-liquidity reservation. The certificate is invalidated by stale or conflicting data, capability loss, early closure, or any state transition outside the registered set.

Proof architecture. The robust clearing theorem uses set inclusion and monotonicity: on one registered operating set, the realized path is bounded below by the proceeds envelope and above by the debt-service envelope. Debt-free finality then becomes a

Table 3: Assumptions and the failure created when each is absent.

Requirement	Consequence when absent
Settled-proceeds lower envelope	Visible depth can overstate repayable cash.
Debt upper envelope	Accrual, fees, or delay can exceed the planned threshold.
Execution authority	The borrower or signer can block hard-flat.
Collateral non-escape	Financed tokens can leave before proceeds repay debt.
Actual tradability through horizon	Scheduled close can be irrelevant after suspension.
Aggregate depth reservation	Multiple positions can reuse the same book.
Settlement-grade evidence	Match or attestation can reduce debt prematurely.
Outcome collateral separation	Residual claim and loan receivable become entangled.

pathwise accounting statement after the stopping time σ . Aggregate results replace independently re-anchored books with one shared execution curve. The impossibility theorem constructs an admissible state in which positive debt remains while the only pledged claim can pay zero and execution or control fails. These arguments require no distributional calibration; practical usefulness does.

Data and code availability

No external dataset is used. The source bundle contains the article, figure assets, and formal checks; the arXiv source history preserves the extended verification materials.

Disclosure

AXIENT is an active research-and-development initiative of the author. Automated tools assisted editorial review, LaTeX support, and deterministic verification. The author specified and reviewed the mechanism, assumptions, formal statements, proofs, configurations, and interpretation, and takes responsibility for the work.

References

- Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3(2): 5–39, 2001. doi: 10.21314/JOR.2001.041.
- Dimitris Bertsimas and Andrew W. Lo. Optimal control of execution costs. *Journal of Financial Markets*, 1(1):1–50, 1998. doi: 10.1016/S1386-4181(97)00012-8.

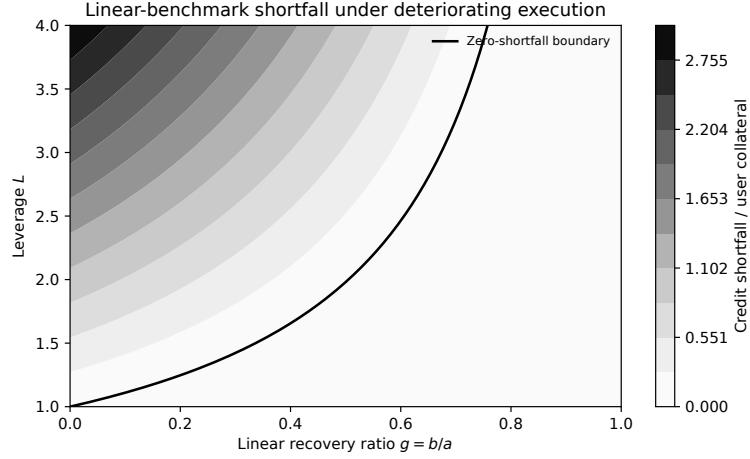


Figure 5: Linear-benchmark shortfall surface as leverage rises and executable recovery falls. Exact certification uses nonlinear shared-book proceeds.

- Dimitris Bertsimas and Melvyn Sim. The price of robustness. *Operations Research*, 52(1):35–53, 2004. doi: 10.1287/opre.1030.0065.
- Rama Cont, Sasha Stoikov, and Rishi Talreja. A stochastic model for order book dynamics. *Operations Research*, 58(3):549–563, 2010. doi: 10.1287/opre.1090.0780.
- Gnosis. Conditional Tokens documentation, release 1.0.3. Protocol documentation, 2020. URL <https://conditional-tokens.readthedocs.io/>. Accessed 13 July 2026.
- Robin Hanson. Combinatorial information market design. *Information Systems Frontiers*, 5(1):107–119, 2003. doi: 10.1023/A:1022058209073.
- Charles F. Manski. Interpreting the predictions of prediction markets. *Economics Letters*, 91(3):425–429, 2006. doi: 10.1016/j.econlet.2006.01.004.
- UMA Protocol. How UMA’s optimistic oracle works. Protocol documentation, 2026. URL <https://docs.uma.xyz/protocol-overview/how-does-umas-oracle-work>. Accessed 13 July 2026.
- Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004. doi: 10.1257/0895330041371321.